

Problem Set 5

Mathematics for Social Scientists

EXERCISE 1. Calculate the following definite and indefinite integrals: (*Some of these are from J. Gill, 2006, Ch5*):

(i) $\int_1^3 (x^2 + 4x + 3) dx$

(ii) $\int_{-1}^1 14 + x^2 dx$

(iii) $\int_2^4 e^y dy$

(iv) $\int (x^2 - x^{-\frac{1}{2}}) dx$

(v) $\int_{-1}^2 x\sqrt{1+x^2} dx$

(vi) $\int_1^2 \frac{1}{t^2} dt$

(vii) $\int e^{3xy} dx$

(viii) $\int 360t^{11} dx$

EXERCISE 2. Solve the indefinite integrals:

(i) $\int \frac{1}{\sqrt{x}} dx$

(ii) $\int e^{-4t} dt$

(iii) $\int x\sqrt{x} dx$

(iv) $\int \frac{1}{x} dx$

(v) $\int (2x^2 + x - 3) dx$

(vi) $\int \frac{(x^4 + 1)^2}{x^3} dx$

EXERCISE 3.

Calculate $\int_0^2 (2x^2 + x - 3) dx$. Hint: Make a sketch of the function before.